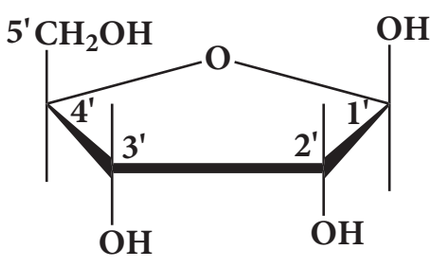
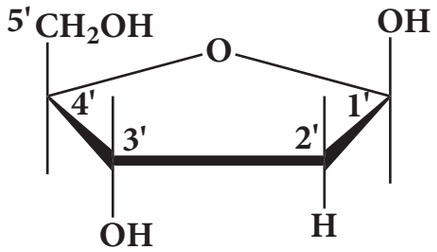
Pentose sugar:

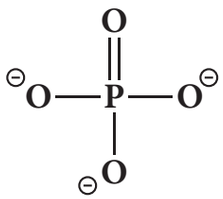
Nucleic acids have two types of pentoses. The recurring deoxyribonucleotide units of DNA contain **2'-deoxy-D-ribose** and the ribonucleotide units of RNA contain **D-ribose**. In nucleotides, both types of pentoses are in their β-furanose (closed five membered rings) form.



Ribose



Deoxyribose



Phosphate group

Phosphoric acid forms phospho diester bond between nucleotides. Based on the number of phosphate group present in the nucleotides, they are classified into mono nucleotide, dinucleotide and trinucleotide.

Nucleosides and nucleotides:

The molecule without the phosphate group is called a **nucleoside**. A **nucleotide** is derived from a nucleoside by the addition of a molecule of phosphoric acid. Phosphorylation occurs generally in the 5' OH group of the sugar. Nucleotides are linked in DNA and RNA by phospho diester bond between 5' OH group of one nucleotide and 3' OH group on another nucleotide.

Sugar + Base ⟶ Nucleoside

Nucleoside + Phosphoric acid ⟶ Nucleotide

nNucleotide ⟶ Polynucleotide (Nucleic Acid)

14.5.2 Double strand helix structure of DNA

In early 1950s, Rosalind Franklin and Maurice Wilkins used X-ray diffraction to unravel the structure of DNA. The DNA fibers produced a characteristic diffraction pattern.

The central X shaped pattern indicates a helix, whereas the heavy black arcs at the top and bottom of the diffraction pattern reveal the spacing of the stacked bases.

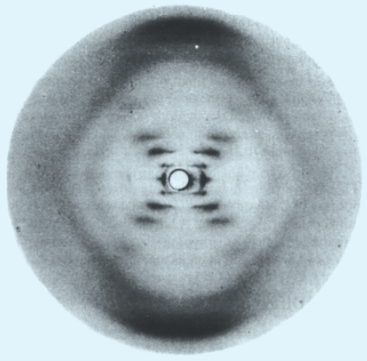


Figure 14.20 DNA X-ray diffraction

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